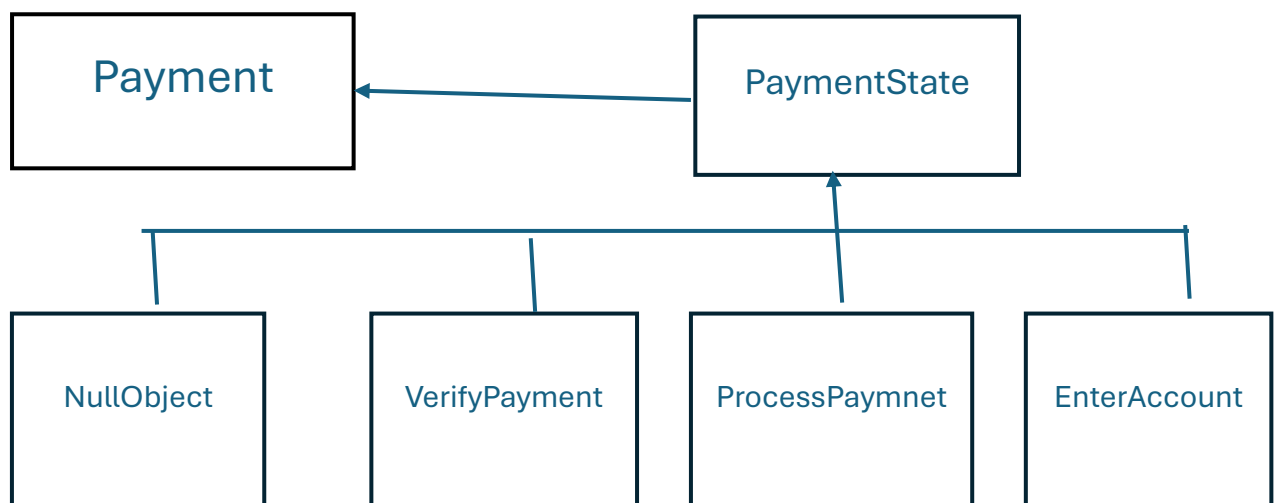


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```

public class Payment
{
    private bool _isProcessing;    // tracks if a payment is already running
    private PaymentState _state;

    public Payment(PaymentState initialState)
    {
        _state = initialState;
    }

    public PaymentState State
    {
        get => _state;
        set => _state = value;
    }

    public void Pay(double amount, string pin)
    {
        if (_isProcessing && !(_state is NullPaymentState))
        {
            // Switch to null-object state if user tries again while busy
            _state = new NullPaymentState();
        }
        else
        {

```

```
        _isProcessing = true;
        _state.Handle(this, amount, pin);
        _isProcessing = false;
    }
}
}
```

```
public abstract class PaymentState
{
    public abstract void Handle(Payment context, double amount, string pin);
}
```

```
public class EnterAccountNo : PaymentState
{
    public override void Handle(Payment context, double amount, string pin)
    {
        Console.WriteLine("Enter account number");
        context.State = new ProcessPayment();
    }
}
```

```
public class ProcessPayment : PaymentState
{
    public override void Handle(Payment context, double amount, string pin)
    {
```

```
        Console.WriteLine($"Processing payment of {amount}...");
        context.State = new VerifyPayment();
    }
}
```

```
public class VerifyPayment : PaymentState
{
    public override void Handle(Payment context, double amount, string pin)
    {
        Console.WriteLine("Verify payment");
        context.State = new OnePaymentAtTime(); // finished or waiting
    }
}
```

```
public class NullPaymentState : PaymentState
{
    public override void Handle(Payment context, double amount, string pin)
    {
        Console.WriteLine("⚠ Only one payment can be processed at a time!");
        // Stay in this state – no further state transitions
    }
}
```